

EGE 2302 PHOTOGRAMMETRY I**DEPARTMENT OF GEOMATIC ENGINEERING AND GEOSPATIAL INFORMATION
SYSTEMS
(GEGIS)****THIRD YEARS 2025/2026 ACADEMIC YEAR****FIRST SEMESTER: - 4th September 2025 – 19th December, 2025****Lecturer** Dr. Charles B. Wasomi (cbwasomi@gmail.com/ 0736170157)**Course outline**

Introduction: Definition, historical development, types of photographs, properties of photogrammetric and remotely-sensed images, electro-optical, linear array, micro-wave systems. **Photographic products:** Orthophotos, photo mosaics. Uses and importance. **Optics:** Newton law, scheimflug and projectivity conditions, aberrations and distortions of images by lenses, depth of object, hyperfocal distance, resolving power. **Aerial mosaics:** Definition, types, uses, construction principles of aerial triangulation. **Aerial cameras:** Types, different components of frame camera accessories, camera calibration, inner orientation, collinearity. **Flight planning:** Purpose and consideration, average scale photography, flight lines, forelap, and overhang design aspects, effects of relief, flight height variations, crab and drift on overlaps, specifications. **Photogrammetry project planning:** Costs, scheduling, specifications, production considerations. **Stereoscopy:** Depth of perception in nature, principles of stereo-viewing, types of stereomodels. **Stereoscopes and their usage:** Principles of instrumentation for analogue and analytical photogrammetry; Types; orientation; relative, absolute. Vertical exaggeration, ground control point selection.

Course Objectives: **Introduce** to the learner Photogrammetry, what it is and its **importance**. Learn on some of the **products of photogrammetry** and their **application**. The learner is to be introduced to the **basic principles** applied in photogrammetry, the **role** of photogrammetry in mapping and data capture.

Be able to **distinguish** the difference between **vertical and tilted** photographs, their **geometric relationship**. Learn on to **determine heights of object** by measurements extracted from aerial photographs.

Have knowledge on photogrammetric products, their methods of production and their application. Learn about **flight planning** process for the purpose of carrying out aerial photography of an area for mapping.

Expectations:

- Understand what Photogrammetry is all about.
- Its **role in mapping** and data capture.
- Have **knowledge on where** photogrammetry is to be applied in mapping of large projects as compared to ground survey.
- Be able to **distinguish** the difference between **vertical and tilted photographs**, their geometric relationship.
- Be able to **determine the heights of object** with the use of aerial photographs **through measurements** to an aerial photograph.
- Have knowledge on some of the photogrammetric products, and the **methods** by which they are produced.
- Have knowledge of the **flight planning** process for the purpose of carrying out aerial photography of an area for mapping.
- Understand on how to create a 3D Model from overlapping photographs and instruments that are used

Methodology:

Lectures; Group discussions; Practical sessions for the students to have some experience on the subject. Encourage questions with explanations and demonstrations offered.

Assignments for group participation during tutorial classes.

Lessons: 6 Hours per week

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| • Tuesday: | 10am-1pm | Practical |
| • Friday: | 1pm - 4pm | Theory/ Tutorial |

COURSE OUTLINE:

- Week 1:** **Introduction:** Definition, historical development, types of photographs, properties of photogrammetric and remote-sensed images. electro-optical, linear array, micro-wave systems.
- Week 2:** **Photographic products;** Othorphotos, photo mosaics. Uses and importance.

 Optics: Newton law, scheimflug and projectivity conditions. Aberrations and distortions of images by lenses, depth of object, hyperfocal distance, resolving power
- Week 3:** **Aerial mosaics:** Definition, types, uses, construction principles of aerial triangulation.
- Week 4:** **Aerial cameras:** Types, different components of frame camera accessories, Camera calibration, inner orientation, collinearity.
- Week 5:** **Continuous Assessment Test (CAT)**
- Week 6:** **Flight planning:** Purpose and consideration, average scale photography, flight lines, forelap, and overhang design aspects.
- Week 7:** effects of relief, flight height variations, crab and drift on overlaps, specifications.

- Week 8:** **Photogrammetry project planning:** Costs, scheduling, specifications, production considerations
- Week 9:** **Stereoscopy:** Depth of perception in nature, principles of stereo-viewing, types of stereomodels
- Week 10:** **Stereoscopes and their usage:** Principles of instrumentation for analogue and analytical photogrammetry.
Types; orientation; interior, relative, absolute. Vertical exaggeration,
- Week 11:** **Ground Control;** point selection
- Week 12:** "
- Week 13:** **CAT**
- Week 14:** **Revision**
- Week 15:** **Semester Examinations.**
- Week 16:** **Semester Examinations.**

Course Evaluation:	C. A. T -	30%
	Practical	15%
	CAT Written	10%
	Assignment	5%
	Semester Examination -	70%

References:

1. Michel Kasser & Yves Egels; Digital photogrammetry.
2. D. E. Murchison; Surveying and Photogrammetry; computation for Civil Engineers.
3. Mather P; Computer precessing of remotely sensed images.
4. Schenk T; Digital Photogrammetry Vol. I.
5. T. J. M. Kennie & G. Petrie; Engineering Surveying Technology.
6. Ramsay J. P. Wilson; Land Surveying.
7. Bannister & S. Raymond; Surveying.
8. W. K. Kilford; Elementary Air Survey.
9. Raymond E. Davis, Francis S. Foote, James M. Anderson, Edward M. Mikhail; Surveying Theory and Practice
10. Internet (Published papers, youtube, Google etc)